Seasonality Anomalies in the US Equity Market

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Introduction

As Halloween approaches, investors revisit a rule of thumb old enough to have entered the language. Sell in May and go away, then return at the end of October. Bouman and Jacobsen (2002) documented the pattern and found that the November to April half-year delivered materially higher returns than the May to October half-year in most of the markets they examined. While McLean and Pontiff (2016) find that return predictability decays substantially after publication, this article asks what remains of these seasonality anomalies in the US equity market and whether trading on them has become less profitable.

The S&P 500 index proxies the US equity market, and the 3-month Treasury bill secondary market rate approximates the risk-free rate. This study examines daily closing levels, their simple returns r_t and the monthly logarithmic returns $R_{y,m}$ of month m in year y from 1981 to 2023.[1]

1 Seasonality Analysis

1.1 Monthly Seasonality

The Halloween effect holds that November to April delivers materially higher returns than May to October. The January effect is the hypothesis that securities prices rise more in January than in any other month, documented for US equities by Rozeff and Kinney (1976).

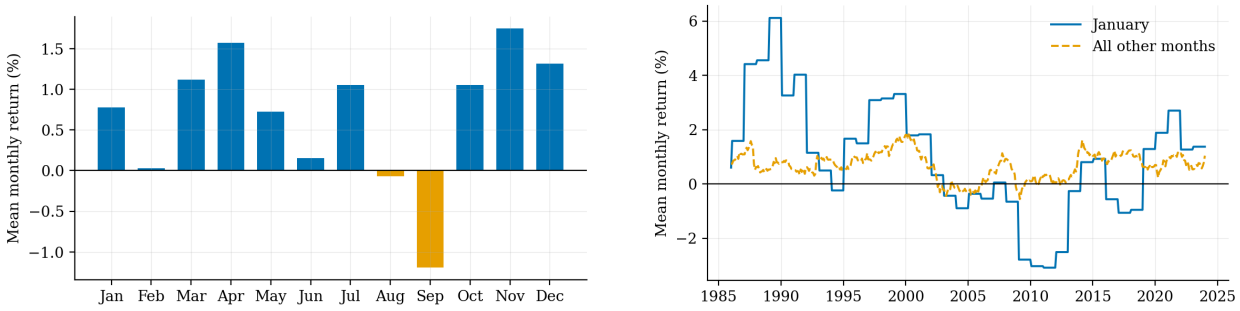


Figure 1. Mean monthly return and the 5-year rolling average of January against all other months.

Figure 1 shows the mean monthly return and the 5-year rolling comparison of January against all other months. January returns on average 0.77% a month, which is slightly above the sample mean. January trails the other months in 48% of the 5-year windows, so the January effect is not stable within this sample.

To investigate monthly seasonality more formally, we write the monthly return as the sum of a yearly level, a seasonal component and a residual, and estimate the seasonal component of month m as the mean deviation of its return from the same year's mean,

$$\hat{s}_m = \frac{1}{|Y_m|} \sum_{y \in Y_m} \left(R_{y,m} - \bar{R}_y \right), \qquad m = 1, \ldots, 12, \tag{1}$$

where $\bar{R}_y$ is the mean monthly return in year y and Y_m is the set of years in which month m is observed.

[1] Code and replication data are available at *GitHub*.

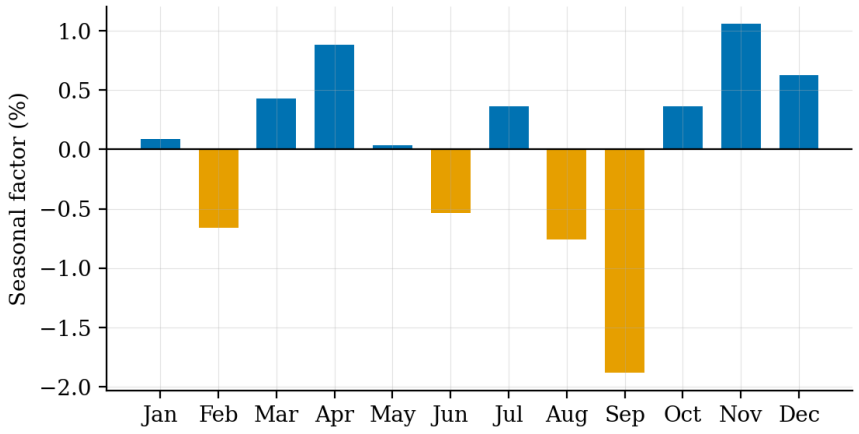


Figure 2. Estimated monthly seasonal factor.

The estimated seasonal factors in Figure 2 support the Halloween effect. November at +1.06% and April at +0.88% are the two strongest months, and both fall inside the November to April window, while September at −1.88% is by a wide margin the weakest and falls outside it. In this sample, the monthly pattern is primarily driven by the weakness of September and the strength of the year-end. Figure 3 compares the two half-years directly. The November to April average exceeds the May to October average by 0.81% per month, with a Newey–West t-statistic of 2.46 with 6 lags (Newey and West, 1987). The gap is negative in only 12.5% of the 5-year windows.

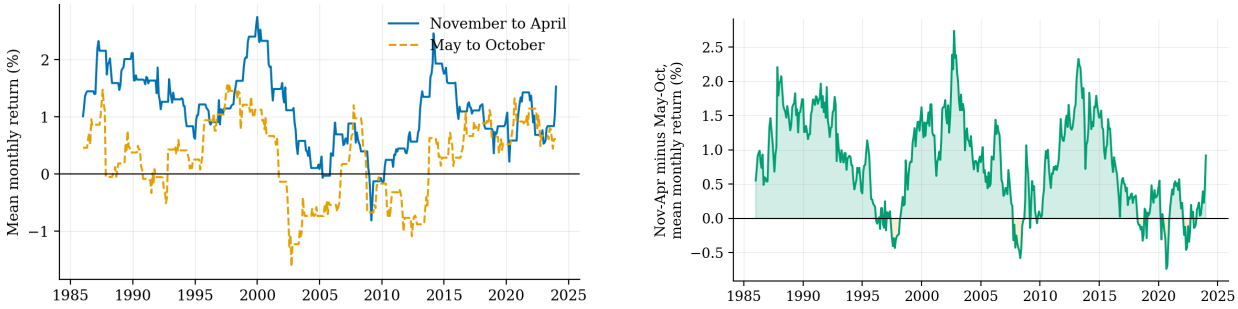


Figure 3. 5-year rolling average monthly return in each half-year, and the difference between them.

1.2 Daily Seasonality

Beyond monthly effects, daily seasonality is also present. The turn-of-month effect holds that returns are abnormally high from the last trading day of a month through the first half of the next (Ariel, 1987). Since months contain between 15 and 23 trading days in this sample, we analyze a forward position j counted from the first trading day and a backward position counted from the last. With $r_{y,m,j}$ the return on trading day j of month m in year y, $n_{y,m}$ the number of trading days in that month and $\bar{r}_{y,m}$ their mean,

$$\hat{s}_j^{+} = \underset{(y,m)}{\text{mean}} \left(r_{y,m,j} - \bar{r}_{y,m} \right), \qquad \hat{s}_j^{-} = \underset{(y,m)}{\text{mean}} \left(r_{y,m,\,n_{y,m}-j+1} - \bar{r}_{y,m} \right). \tag{2}$$

Figure 4 reports the seasonality of the first and last 5 trading days. The first trading day carries a factor of +0.147% and the second +0.061%, and only the first is significant at the 5% level ($t = 2.82$). At the other end of the month, the fourth, third and second to last are positive and the final trading day is −0.010%, none significant at the 5% level.

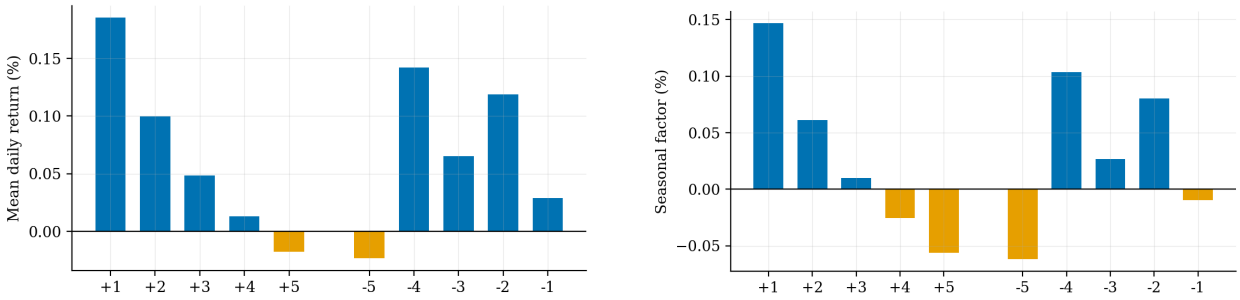


Figure 4. Mean daily return and estimated daily seasonal factor over the first and the last 5 trading days.

1.3 January Barometer

The January barometer asserts that the January return predicts the return over the remainder of the year (Cooper et al., 2006). The Pearson correlation between the January return and the February-to-December return is +0.156 across 43 years ($t = 1.01$, $p = 0.32$). Figure 5 shows that the 5-year rolling Pearson and Kendall correlations change sign repeatedly. Hence, the data provide no evidence for the barometer.

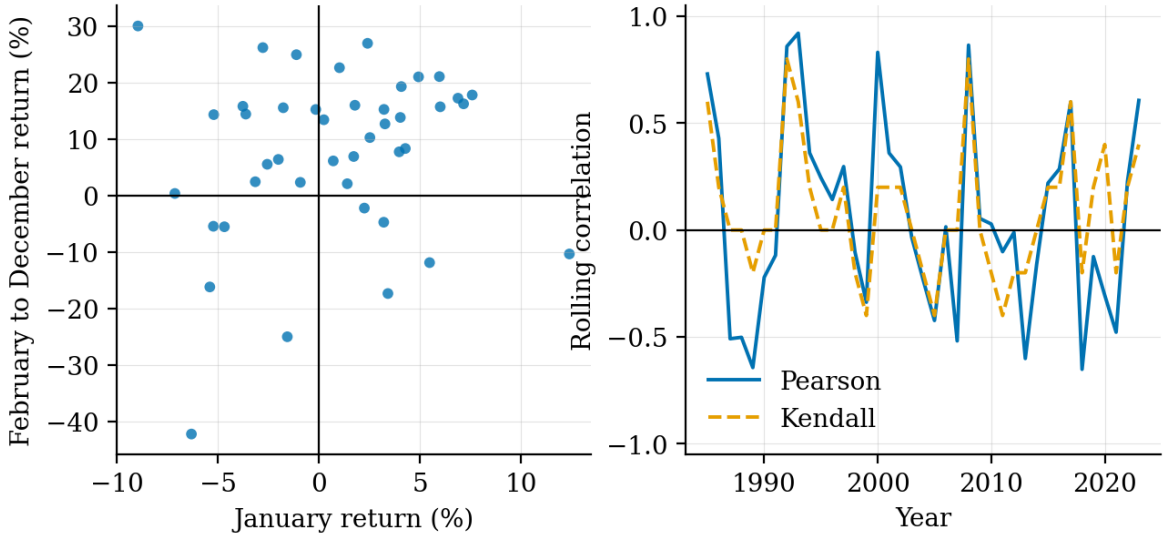


Figure 5. January return against the February to December return, and their 5-year rolling Pearson and Kendall correlations.

2 Backtest Result and Discussion

2.1 Trading Strategies

S1 Halloween Strategy One buys at the close of the last trading day of October and sells at the close of the last trading day of April, holding Treasury bills for the remainder of the year.

S2 Halloween and Turn-of-Month Strategy On top of S1, one also holds the index over the turn-of-month window, the first and the last five trading days of each calendar month. Where the two signals coincide, the position is levered to 1.5 with the additional half funded at the bill rate.

BH Buy and Hold One buys the index at the start of the sample and holds it throughout. The position is credited with the sample average annual dividend yield of 2.48% accrued evenly across

trading days, and S1 and S2 earn the same yield while they hold the index.

2.2 Backtest Result

Strategy	Return (%)	Volatility (%)	Sharpe	Sortino	Max DD (%)
S1 Halloween	9.69	12.53	0.457	0.649	−35.86
S2 Halloween and turn-of-month	12.81	17.87	0.494	0.704	−46.28
BH Buy and hold	10.75	18.06	0.376	0.520	−55.24

Table 1. Performance from 1981 to 2023. Return is the annualized mean log return, volatility is annualized and Max DD is the maximum drawdown.

The backtest charges 5 basis points (bps) one-way on every change in position. S2 outperforms BH on every measure in Table 1, and its maximum drawdown is 9% shallower. S1 trails BH on return, but its much lower volatility lifts its Sharpe ratio above that of BH.

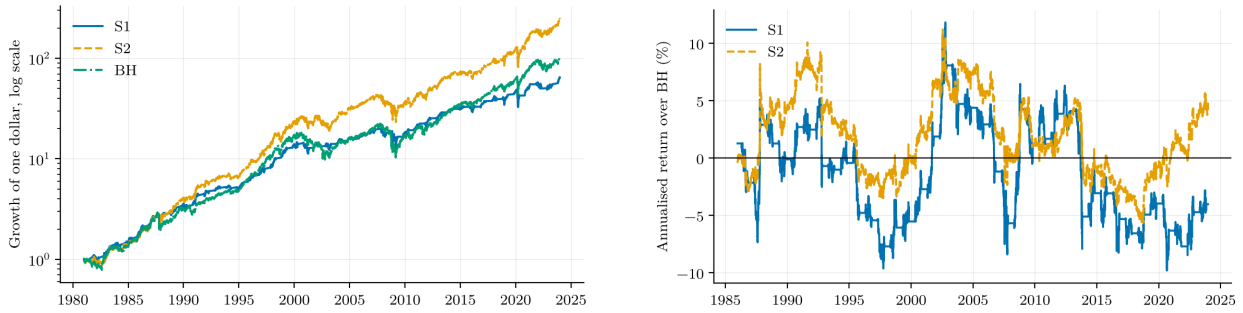


Figure 6. Growth of one dollar under S1, S2 and BH, and the 5-year rolling outperformance of S1 and S2 over BH.

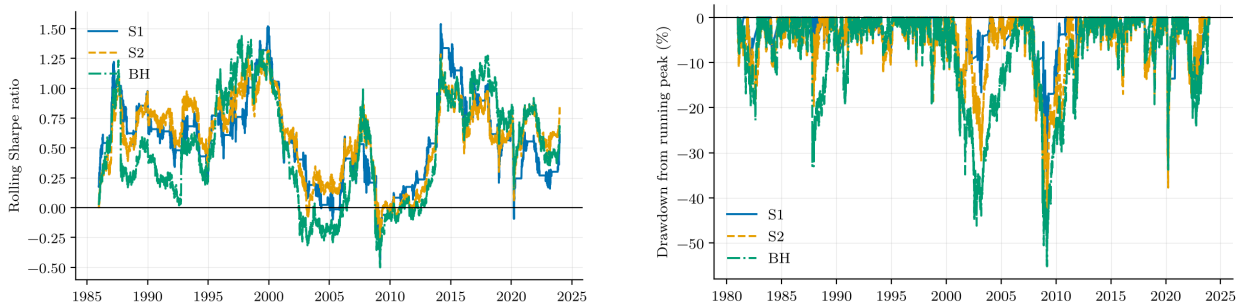


Figure 7. 5-year rolling Sharpe ratio, and drawdown from running peak.

Figure 6 traces the strategies against BH over time. Over the non-overlapping periods 1981 to 1999, 2000 to 2019 and 2020 to 2023, S2 outperforms BH by +1.74%, +1.79% and +4.96% a year, and a linear trend fitted to its annual outperformance has a slope that is not significant ($t = -0.25$). Over the full sample, its outperformance of 2.06% a year is not statistically significant either (Newey–West $t = 1.36$ on the daily differences with 10 lags). Figure 7 turns to risk. The Sharpe ratio of S2 leads that of BH by 0.150, 0.095 and 0.176 over the same periods, so the lead does not narrow over time. The drawdown panel shows S1 is shallower than BH on 65% of days and S2 on 52%, with the gap widest after the dot-com crash in 2002 and, for S1, in the global financial crisis of 2008.

3 Discussion

In this study, we analyze monthly and daily seasonality in the US equity market, backtesting the Halloween (Sell in May and go away, then return at the end of October) strategy and the turn-of-month effect over the period of 1981 to 2023. S2 outperforms BH on every measure, but its excess return is not statistically significant and shows no decline over time.

Hence, we cannot conclude that trading seasonality anomalies earned an excess return in the past, nor that this excess return has decayed. Our study did not capture the effect of technological advancement. Today, every data vendor offers real-time or intraday price feeds, corporate calendars, and economic releases, whereas a hedge fund two decades ago might have waited days to receive cleaned data. Meanwhile, trading infrastructure has transformed beyond recognition. Electronic trading platforms and direct API access to exchanges have driven both commissions and latency to historic lows. An active fund in the 1980s might have paid higher commissions and waited minutes for a fill, while retail platforms today can charge zero commissions and execute orders in milliseconds. Our backtest, charging 5 bps and neglecting other frictions such as slippage, market impact and taxes, might not reflect the reality of trading in the earlier decades. Understating historical costs inflates the early-period outperformance and therefore inflates any apparent decay. The constant dividend yield works in the same direction. S1, S2 and BH all earn 2.48% a year while holding the index, whereas the actual yield was higher before 2000 and lower after, so the dividends that S1 and S2 forgo out of the index are understated early in the sample.

3.1 Moving Forwards

Our analysis has focused exclusively on the S&P 500, which likely understates the full extent of calendar-based anomalies. Small- and mid-cap stocks, as well as many international markets, may exhibit stronger seasonality due to lower liquidity, narrower analyst coverage, and higher access barriers. On the modeling side, our seasonal decomposition might be too naive, and a more robust framework, such as a Bayesian Structural Time Series model, could potentially estimate and forecast the seasonality factor better. Furthermore, incorporating variables such as aggregate fund flows, liquidity cycle indicators, shifts in investor sentiment, tax-loss-harvesting windows, monetary policy transitions and recessionary episodes into a multivariate setup could identify hidden patterns beyond seasonality.

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